

Cheat Sheet - Algorithm and Programming (First Meeting)

What is an Algorithm?

An algorithm is a logical, step-by-step procedure to solve a problem. It's like a recipe or a list of instructions that anyone (or a computer) can follow to achieve a goal.

Difference Between Algorithm, Programming, and Program:

Algorithm = the plan or steps.

Programming = the process of writing code.

Program = the final working product.

Algorithm = the idea.

Programming = turning the idea into code.

Program = the finished product that runs.

Why Computers are Literal:

Computers follow instructions exactly as given. They do not assume or guess. Every single step must be super clear. If anything is missing or vague, ERROR happens.

Example: If you tell a computer "make tea" without steps, it will not know what to do. You must define every tiny action (boil water, add tea, etc.).

Baby Analogy:

Think of a computer like a baby who knows NOTHING. You must teach it from zero:

1. Move your hand.
2. Grab the toy.

3. Lift your hand.
4. Shake it.
5. Smile.

That's why algorithms and programming need detailed, clear, and logical steps.

First Meeting Recap:

For the first class, we only covered the basics above to understand the core concepts.